

# Aditya Shirsatrao

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Solapur, Maharashtra, India

## Objective

Software engineering undergraduate with strong fundamentals in distributed systems, data structures, and object-oriented programming, and hands-on experience building low-latency, event-driven backends at scale. Seeking Goldman Sachs's Software Engineering Summer Internship to contribute to high-performance financial technology systems in a collaborative, risk-focused environment.

## Education

N. K. Orchid College of Engineering & Technology Solapur, Maharashtra  
Bachelor of Technology in Artificial Intelligence and Data Science – CGPA: 8.5/10 Aug. 2023 – Present (Expected 2027)  
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, OOP, Software Engineering, Machine Learning, Computer Networks.

## Experience

Fullstack Web Development Intern Unified Mentor Pvt. Ltd  
Distributed Systems & Backend Engineering Jul. 2025 – Jan. 2026  
Designed a cloud-native microservices e-commerce platform with payment orchestration, inventory consistency, and distributed caching using an event-driven architecture on Docker and Kubernetes.  
– Built an AI-powered code review platform processing GitHub pull requests asynchronously; integrated Kafka message queues and Redis caching, reducing analysis latency by 45% via parallel distributed workers.  
– Engineered a real-time collaborative IDE with sub-100ms synchronization using WebSockets, Socket.IO, Redis pub/sub, and CRDT-based operational transforms.  
– Developed an event streaming analytics system with Kafka and Apache Spark Streaming for near real-time dashboards, anomaly detection, and alerting.  
AI/ML Lead Google Developer Groups on Campus (GDG-NKOCET)  
Leadership & Mentorship Sep. 2025 – Present  
Lead technical workshops and hackathon mentorship for a 200+ student community, emphasizing clean architecture, testing, and system design.

## Projects

Fraud Detection System | FastAPI, Autoencoders, Redis, React | Code Dec. 2024 – Present  
– Architected a real-time anomaly detection engine on 500k+ transactions; optimized inference to <50ms with async I/O, supporting 100 requests/second.  
– Reduced manual fraud-review workload by 40% via a visualization dashboard.  
DevGuard – Role-Based Access Control APIs | Express.js, JWT, PostgreSQL, Prisma Oct. 2024 – Nov. 2024  
– Built a secure backend API gateway with RBAC and multi-session token logic for enterprise applications.  
– Designed the Prisma schema, modular auth middlewares, and an endpoint access matrix.  
Chess Engine | C, Minimax with Alpha-Beta Pruning Jun. 2024 – Aug. 2024  
– Engineered an AI-driven chess engine with custom board evaluation heuristics, move ordering, and depth-limited search; optimized decision speed in critical positions.  
Backend ML Services Framework (Core-4) | Python, FastAPI, Design Patterns Feb. 2026 – Mar. 2026  
– Developed a modular backend framework using factory and strategy design patterns, reducing code duplication by 35% and achieving >80% test coverage.

## Technical Skills

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL  
Backend & Distributed Systems: FastAPI, Node.js, Express.js, Redis, Kafka, Message Queues, Microservices, REST APIs, WebSockets  
Infrastructure: Docker, Kubernetes, GitHub Actions CI/CD, AWS (EC2/S3/Lambda), Linux, Grafana/Prometheus  
Core Concepts: DSA (366+ LeetCode), System Design, OOP, Concurrency & Asynchronous Programming, Databases, Security (RBAC, JWT)

## Achievements & Certifications

Competitive Programmer: 366+ LeetCode problems (104 Easy, 176 Medium, 86 Hard) + 50 SQL problems.  
IEEE First-Author Publication at ICCTWC 2026; 1st Place Hack to Future 3.0 (600+ teams); 1st Runner-Up Orchathon 2K26 (800+ teams).  
Open Source: 50+ PRs merged across Google (OSS-Fuzz), Microsoft (VibeVoice), Apache (DataFusion), Hedera Hashgraph (Hiero SDK).  
Certifications: JPMorgan Chase & Electronic Arts SWE Virtual Experience; NPTEL Programming in Java (Elite+Gold, 91/100); Kaggle AI Agents Intensive Course.  
Leadership: Class Representative for a 120+ student cohort; seminar speaker on AI and Chandrayaan-3.